

Paper A v0.9 — The Physical (SU(3))
 Yang–Mills Mass Gap via D_{IV}^5 Spectral
 Geometry. Companion to Paper B. v0.9 = the
 gap value CLOSED (K1711). Reading the field
 correctly — Yang–Mills is the gauge field, a
 1-FORM (the adjoint) — fixes the operator as
 the FORM (Hodge) Laplacian, not the scalar
 one. Forms on S^4 start at $2(n_C-2) = 2N_c = 6$
 (the color/adjoint eigenvalue, not the scalar's 4;
 passes the dimension-generic test —
 $2(n_C-2)=6$ requires $\dim=n_C-1=4$). The
 form parity ($k+p+m$ even) KEEPS the
 pure-spatial (1,0) mode, and gauge invariance
 eats A_θ (pure gauge for $m \neq 0$), so the physical
 transverse gluon is (1,0), $m=0 \rightarrow \text{mass}^2 = 2N_c$
 $= 6$, R-INDEPENDENT, color-rooted — a
 DERIVED value on the compact boundary.
 LABEL DISCIPLINE (Guard 3): it is $2N_c$,
 NEVER ' $= C_2$ ' (the scalar Casimir n_C+1 —
 the morning's identity, K1707; same number,
 different object). The scalar readings (4, the
 §570 (1,1) candidate) were the wrong sector,
 superseded by the form reading.

Polyakov/Wilson-line mode at 10 above the gap
 (deconfinement tie). TWO NAMED
 RESIDUALS: (a) the $\Delta \rightarrow \text{mass}^2$ shift (Elie); (b)
 the flat- $\mathbb{R}^4$ construction (Clay). Glueball 1720
 separate-and-valid (on C_2^{int}), out of the room.
 The package intro can move now — the value
 is closed. NOT ‘YM proved’, NOT ‘ $= C_2$ ’. Count
 HOLDS 4.

Lyra (Claude Opus 4.8) — Casey Koons, PI; Grace (OS reduction + T2490/T2491 spectral casc

2026-08-19 Wednesday (date-verified)

The Physical Yang–Mills Mass Gap via D_IV⁵ Spectral Ge- ometry

Paper A of the two-paper package (companion: Paper B, substrate uniqueness). v0.9 — the gap value is CLOSED (K1711): the gauge field is a form, so the operator is the form (Hodge) Laplacian, whose lowest S^4 eigenvalue is $2N_c = 6$; the pure-spatial (1,0) gluon gives $\text{mass}^2 = 2N_c = 6$, R -independent, color-rooted — derived. Labeled $2N_c$, never “ $= C_2$ ”. Two named residuals: the $\Delta \rightarrow \text{mass}^2$ shift, and the flat- $\mathbb{R}^4$ construction. Glueball separate-and-valid, out of the room. The package intro can now move.

0. Two guards and one residual (read this first)

Two objects must not be fused, and one residual must not be dressed as a proof.

- Guard 1 (space). The gap we derive is a spectral gap on the compact boundary of D_IV⁵ — a *different operator on a different space* from the Yang–Mills Hamiltonian on flat $\mathbb{R}^4$. $\text{mass}^2 = 2N_c = 6$ on our boundary is not, by assertion, the $\mathbb{R}^4$ mass gap; the two are connected only through a bridge that must be exhibited, never fused (the standing K932 guard: exhibit the bridge, don’t assert a shared eigenvalue). (*This is residual (b), the flat- $\mathbb{R}^4$ construction.*)
- Guard 2 (object). The pure-Yang–Mills (Clay) gap is the glueball (1720 MeV, 0.6% vs lattice), not the proton (938 MeV, which is the *full-QCD*, matter-included gap — a different object); we do not lend the proton’s four-decimal precision to the Yang–Mills gap (K1705). And the glueball

is *out of the room* for the *gap-value* pin (K1708/K1709): $1720 = (11/6) \cdot 938$ is built on C_2^{int} (the labeled integer primary), not on λ_1 (the boundary Laplacian gap), and its $11/6$ carries $C_2 = 6$ — so it is a real, separate forward prediction but not independent evidence for the gap value. The value is pinned from the geometry alone (the GATE-KOIDE discipline that keeps $Q = 2/3$ out of the Koide pin).

- Guard 3 (the symbol “ C_2 ” itself — K1707). Three distinct objects all read “ C_2 ” and all equal 6 at the physical point $n = n_C = 5$, so writing “the gap = $C_2 = 6$ ” hides *which one* is meant:
 - $C_2^{\text{int}} := n_C + 1 = 6$ — the *labeled integer primary* (one of the five integers). A definition.
 - $C_2^{\text{flow}}(n) = 2n - 4$ — the *commitment-flow Casimir* (a function of n ; = 6 at $n=5$ only by the coincidence-at- $n_C=5$).
 - $C_2^{\text{Berg}} = n + 1$ — the *Bergman-representation Casimir* (a function; = 6 at $n=5$, and equal to C_2^{int} *identically*, since C_2^{int} is *defined* as $n+1$).
 The consequence (Cal, K1707): “ $\lambda_1 = C_2 = 6$ ” as previously written is $C_2^{\text{int}} = C_2^{\text{Berg}}$ — the same definition on both sides, an identity that *cannot fail* ($P^2=P$), not a derived tower-gap. The genuine, falsifiable question is whether the *physical mass-gap operator’s* tower-gap equals 6 non-tautologically — which awaits pinning the one operator from the primary source (Section 2).
- The one residual (flat space). The honest open problem is a single question: is the theory that the reconstruction machinery (Rehren holography $\rightarrow$ Osterwalder–Schrader) places on the flat 4D boundary literally interacting $SU(3)$ Yang–Mills, or a gapped generalized-free cousin that also satisfies the axioms? This is the large named open — carried as such, never a proof.

The result, and the two residuals — in one sentence. On the compact boundary the Yang–Mills mass gap is a derived value — $\text{mass}^2 = 2N_c = 6$, color-rooted and R-independent (the gauge field is a form; forms on S^4 start at $2(n_C-2) = 2N_c$; gauge invariance eats the time-component, leaving the pure-spatial (1,0) gluon) — with two named residuals: (a) the $\Delta \rightarrow \text{mass}^2$ shift (whether the conformal weight Δ shifts the Laplacian-eigenvalue $\rightarrow$ physical-mass² identification), and (b) the flat- $\mathbb{R}^4$ construction (that this compact-boundary gap is the Clay $\mathbb{R}^4$ Yang–Mills gap).

So the paper’s claim is explanatory, and now with a closed value: BST *derives* the Yang–Mills mass-gap value on its compact boundary ($2N_c = 6$), with the flat-space construction the named residual. The gap exists because the domain is bounded, its value is fixed by the geometry (no dimensionless knob), and the number is color-rooted ($2N_c$) — not the scalar Casimir C_2 (Guard 3: they coincide at 6 but are different objects; “= C_2 ” was the morning’s identity). Nothing here is “we solved Yang–Mills”: residual (b) is the large flat- $\mathbb{R}^4$ open, carried as such.

Abstract

We study the Yang–Mills mass gap for the physical gauge group $SU(3)$ — color — derived (not assumed) from the substrate D_{IV}^5 in the companion Paper B. The mass-gap operator is a Laplacian on the physical Shilov boundary $\partial_S D_{IV}^5 =$

$(S^4 \times S^1)/\mathbb{Z}_2$; its gap is nonzero because the domain is bounded, with no free dimensionless parameter beyond the ruler R . The value is derived (K1711): $\text{mass}^2 = 2N_c = 6$, color-rooted and R -independent. The decisive step is reading the field correctly — Yang–Mills is the gauge field, a 1-form (the adjoint), so the operator is the form (Hodge) Laplacian, whose lowest eigenvalue on S^4 is $2(n_C - 2) = 2N_c = 6$ (not the scalar’s 4; a dimension-specific, color-rooted number). On $(S^4 \times S^1)/\mathbb{Z}_2$ the form parity ($k+p+m$ even) keeps the pure-spatial mode $(1,0)$, and gauge invariance eats the time-component (A_θ pure gauge for $m \neq 0$), so the physical transverse gluon is $(1,0)$, $m=0$ — R -independent. This 6 is $2N_c$, the color/adjoint eigenvalue, and must not be relabeled “ $= C_2$ ” (the scalar Casimir n_C+1 , a different object; equating them was the identity trap corrected earlier this session, K1707). The earlier scalar readings (bare- S^4 “4”, the $\mathbb{Z}_2$ -projection, the §570 (1,1) candidate) were the wrong sector, superseded by the form reading. Two named residuals remain: (a) the $\Delta \rightarrow \text{mass}^2$ shift (whether the conformal weight Δ shifts the eigenvalue $\rightarrow$ physical-mass²; if $\text{mass}^2 = 6$ directly, the close hardens), and (b) the flat- $\mathbb{R}^4$ construction (the Clay identification). (*The glueball 1720 MeV = (11/6)·938 is a separate valid prediction built on C_2^{int} — kept, but out of the room as gap-evidence, K1708.*) The construction does not build 4D Yang–Mills from scratch on flat $\mathbb{R}^4$; instead the D_{IV}^5 spectral theory directly supplies Osterwalder–Schrader reconstruction data (a rigorous Hilbert space, a gapped Hamiltonian, reflection positivity, 4D conformal/Poincaré covariance, clustering) on the domain’s 4D conformal boundary ($SO(4,2) \subset SO(5,2)$), so the existence of a rigorous gapped 4D theory follows from classical harmonic analysis on a bounded symmetric domain. The problem then reduces to a single residual (Section 6, Section 9): is this gapped edge theory genuinely interacting Yang–Mills, or a (also-gapped, also-axiom-satisfying) generalized-free theory — equivalently, is the Rehren-reconstructed boundary theory *literally* $SU(3)$ YM? In v0.4 we report that the decisive probe — the cross-channel glueball spectrum — has been computed and materially advances the interacting reading: the four ground J^{PC} channels follow a linear conformal-energy ladder off one verified number (the genus of D_{IV}^5), with one blind leg, $2^{++} = g/n_C$, and the other channels consistent within current lattice error. A generalized-free theory produces no such bound-state J^{PC} tower — real evidence on the interacting side — but the interacting-vs-free question remains the named-open residual. We do not claim the Clay problem is solved. The honest statement is a decisive *sharpening* to one residual, with the flat- $\mathbb{R}^4$ identification carried as the large named open.

1. Scope: the physical theory, not “all G ”

The Clay problem asks for a mass gap for *every* compact simple gauge group. BST addresses the physical theory: $G = SU(3)$, color. The companion Paper B proves D_{IV}^5 is the unique substrate and that $SU(3)$ is derived, not chosen. So “all G ” is not BST’s claim. This paper makes claims only about the physical theory and is explicit about that scope (Section 9).

2. The mass gap: value DERIVED on the compact boundary — $\text{mass}^2 = 2N_c = 6$, color-rooted, R-independent (K1711); two named residuals

What is SOLID (existence). The mass-gap operator is a Laplacian on the Shilov boundary of D_{IV}^5 — a positive operator whose spectrum *is* the arrow of time. The domain is bounded, so the spectrum is discrete and bounded below; the gap is nonzero *because the domain is bounded* (decompactify, $R \rightarrow \infty$, and the gap $\rightarrow 0$), with no free dimensionless parameter beyond the ruler R (a Planck-unit — the same single input GR takes as G and QM as $\hbar$). *Existence is not in question and never was.*

The value closes once you ask what the field is (K1711). The morning's confusion came from treating the gap as a *scalar* Laplacian eigenvalue. But Yang–Mills is the gauge field = the adjoint, a 1-form — so the operator is the form (Hodge) Laplacian, not the scalar one. Three facts then close the value:

1. Forms on S^4 start at $2(n_C - 2) = 2N_c = 6$ — not the scalar's 4. The lowest eigenvalue of the Hodge Laplacian on co-closed 1-forms of S^4 is $2N_c$, a color-rooted number (the adjoint, $2(n_C - 2)$), and it passes the dimension-generic test: $2(n_C - 2) = 6$ requires $\dim = n_C - 1 = 4$ specifically — it is *our* dimension's value, not a coincidence of a formula that gives 6 at every rank.
2. The form- $\mathbb{Z}_2$ keeps the pure-spatial mode. On $(S^4 \times S^1)/\mathbb{Z}_2$ the identification is $k + p + m$ even (spatial level k , form degree p , time mode m). For the gauge 1-form ($p = 1$) the mode $(1, 0)$ has $k + p + m = 2$, even — kept. The form degree supplies the parity that the scalar lacked; nothing is projected out.
3. Gauge invariance eats the time-component. The A_θ (time-circle) component is pure gauge for $m \neq 0$, so it carries no physical mode; the physical transverse gluon is $(1, 0)$, $m = 0$, pure-spatial. Hence the gap is R-independent — no time-circle radius enters.

$\Rightarrow \text{mass}^2 = 2N_c = 6$, R-independent, color-rooted — a derived value on the compact boundary.

Label discipline (Guard 3, load-bearing — K1711). This 6 is $2N_c = 2(n_C - 2)$, the color/adjoint form-eigenvalue. It must never be relabeled “= C_2 .” $C_2 = n_C + 1$ is the *scalar* Casimir, a different object that happens to equal 6 at $n_C = 5$; equating them was exactly the morning's identity trap (K1707). Write $2N_c$; the gap is color-rooted, not the scalar Casimir.

A check above the gap. The Wilson-line / Polyakov (holonomy) mode sits at 10, above the gap — consistent with the confined phase and tying to the deconfinement scale (Section 7 / the T_c work). The gap is the lowest, at $2N_c$.

How the earlier readings resolve (audit trail). The scalar “4” (bare S^4), the $\mathbb{Z}_2$ -projection of it, and the §570 spatial-mode candidate $(1, 1) = 4 + 1/R^2$ were all the scalar sector — the wrong sector for a gauge field. Reading the field correctly (a form) supersedes them: the form $(1, 0)$ mode is pure-spatial and R-independent, so

the R-dependence and the “which $\mathbb{Z}_2$ -even mode” question both evaporate. The old “6” (conformal reading = $C_2 = n_C + 1$) and “7” (genus) were never the gap.

reading	value	status
scalar S^4 Laplacian, k(k+3) first level	4	wrong sector (scalar); $\mathbb{Z}_2$ -projected-out anyway
scalar §570 spatial-carrying mode (1,1)	$4 + 1/R^2$	wrong sector; superseded by the form reading
conformal reading / Bergman-rep Casimir, n+1	6	= C_2 — the scalar identity (K1707), not the gap
genus, $g = n_C + 2$	7	a different invariant
gauge-form (1,0), Hodge Laplacian	$2N_c = 6$	the gap — color-rooted, R-independent, DERIVED (K1711)

The glueball stays separate and out of the room. The glueball 1720 MeV = $(11/6) \cdot 938$ is a real forward prediction (Section 7) built on C_2^{int} (the labeled integer), not on the boundary gap $2N_c$ — so it cannot corroborate the gap value (GATE-KOIDE). Kept as a prediction; out of the room for the pin.

Tier: gap VALUE = DERIVED on the compact boundary — $\text{mass}^2 = 2N_c = 6$, color-rooted, R-independent (K1711). Two named residuals remain (stated up front, Section 0): (a) the $\Delta \rightarrow \text{mass}^2$ shift — whether the conformal weight Δ shifts the Laplacian-eigenvalue $\rightarrow$ physical-mass² identification (Elie owns; if $\text{mass}^2 = 6$ directly, the close hardens); (b) the flat- $\mathbb{R}^4$ construction — that this compact-boundary gap is the $\mathbb{R}^4$ Yang–Mills gap (the Clay residual, Sections 6/9). *Guard 1 holds: this is the gap of a boundary operator on D_{IV}^5 ; the $\mathbb{R}^4$ identification is residual (b), never fused.*

3. W4 dissolved: the gauge group is derived (Paper B)

A referee objection — “you treat only SU(3)” — is answered by showing G is not a free input. Paper B establishes D_{IV}^5 as the unique irreducible Hermitian symmetric domain meeting prior, dimension-innocent criteria, forcing $\dim_C = 5$ and $N_c = 3$ (the dimension-free output of the proved T1829). $N_c = 3$ is the T1829 output (a value-recurrence, Cal #335); the su(3)/color readings of “3” are *downstream of the open color identification*, not independent derivations. So SU(3) is substrate-derived. The whole dynamical content cascades from one physical input — three colors fix $\text{rank} = 2$ ($N_c = \text{rank}^2 - 1$), and rank generates $\{n_C, C_2, g\}$ (T2491). The “all G” wall is dissolved.

4. W1 folded: the D_{IV}^5 spectral theory supplies the OS data on the 4D conformal boundary

BST does not construct 4D YM from scratch on $\mathbb{R}^4$. The OS reconstruction theorem builds a QFT from a data set, and the D_{IV}^5 spectral theory carries it — on the domain's 4D conformal boundary, which exists because D_{IV}^5 carries the 4D conformal group $SO(4,2) \subset SO(5,2)$ (the same edge where light and electromagnetism live):

OS datum	D_{IV}^5 realization	tier
rigorous Hilbert space	the Hardy space $H^2(D_{IV}^5)$	SOLID
Hamiltonian with a gap	form (Hodge) Laplacian on $(S^4 \times S^1)/\mathbb{Z}_2$; gauge-form (1,0) pure-spatial; gap = mass ² = $2N_c = 6$, color-rooted, R-independent (K1711)	DERIVED (residuals: Δ -shift, $\mathbb{R}^4$ construction)
reflection positivity	tube-type (Cayley $\rightarrow T(\Omega)$); Θ = cone involution; OS-RP (free level) = Cauchy–Szegő RKHS-positivity	SOLID (free level)
4D conformal/Poincaré	$SO(4,2) \subset SO(5,2)$, UV-conformal/IR-gapped	SOLID-structural
clustering / unique vacuum	from the spectral gap $\Delta > 0$ (nonzero, geometry-fixed)	SOLID

So a rigorous gapped 4D theory follows from classical harmonic analysis; the open from-scratch construction is sidestepped, not solved. The single non-free-level piece is the interacting upgrade of reflection positivity — which is the residual (Section 6). Tier: SOLID at free level; interacting-RP = the residual.

5. Net-compatibility via Bisognano–Wichmann (SOLID-CONDITIONAL)

The HS (Hardy–Szegő) isometry intertwines the bulk and boundary operator nets: HS is $SO(5,2)$ -equivariant, and both the bulk Rehren net and the boundary net are modular reconstructions of the *same* $SO(5,2)$ positive-energy representation via Bisognano–Wichmann (BGL) / Borchers. So locality/causality transfer across HS. Tier: SOLID-CONDITIONAL on the BGL hypothesis-checks. (*This is the modular-geometry rung of the reconstruction stack named in Section 9.*)

6. The one residual: interacting Yang–Mills, or generalized-free? (= “is the Rehren edge literally YM?”)

The prize is a single question, and F1056 and v0.3 name it two ways that are one residual:

Is the OS/Rehren-reconstructed gapped 4D boundary theory genuinely interacting SU(3) Yang–Mills, or a generalized-free gapped theory?

The two namings coincide: - v0.3’s naming (interacting vs generalized-free): a generalized-free field also satisfies the OS axioms and also has a gap — so the residual is to show the reconstructed theory is genuinely *interacting*. - F1056’s naming (Rehren-edge = literally-YM): Rehren’s algebraic holography is a rigorous bulk $\leftrightarrow$ boundary net correspondence, but its *dual is not automatically the physical CFT* — so the residual is to show the Rehren edge theory is *literally* SU(3) YM, not a cousin.

These are the same open piece. “Genuinely interacting” and “literally the physical YM net” are one condition, viewed from the OS side and the Rehren side respectively.

A trap we avoid. “Non-commutative $\implies$ interacting” is false — a free field also has a non-commutative CCR algebra. Interaction means nonzero connected n -point functions ($n \geq 3$). The YM self-interaction is the vertex $[A,A]$, nonzero precisely because the gauge algebra is non-abelian. So

interacting $\iff$ the gauge algebra is genuinely non-abelian $\mathfrak{su}(3) \iff$ the cross-channel glueball spectrum matches SU(3) (Section 7).

Two pieces of evidence toward the interacting side: - The bulk-color algebra closes as non-abelian $\mathfrak{su}(3)$ (bilinear Schwinger realization; Elie 4301). The substrate identification (#418) is in progress. - The spectrum is non-additive and computed (Section 7). A generalized-free theory has an additive multi-particle spectrum and no genuine bound-state J^{PC} tower. The computed ladder is non-additive and channel-structured — evidence of binding. This is *evidence toward* interacting, not a closure of it.

7. The cross-channel glueball spectrum — named-open, materially advanced

In v0.1 this section was *named-open at structural tier*. Since v0.3 it is computed and materially advanced — still named-open (the honest tier, after Cal’s trim of the v0.2 over-claim “derived / landed interacting”).

7.1 Why there are exactly four channels (operator-algebraic closure)

The gauge field F is a rank-2 antisymmetric tensor operator on $H^2(D_{\text{IV}}^5)$. Its bilinears $F \otimes F$ decompose into exactly four irreducible J^{PC} components, each of which must have eigenstates for the substrate’s operator algebra to close:

channel	operator	what it commits
0^{++}	$\text{Tr}(F^2)$	energy density (the action)
0^{-+}	$\text{Tr}(F \tilde{F})$	topology (Pontryagin density)
2^{++}	$\text{Tr}(F^{\wedge\mu\rho} F^{\wedge\nu}_{\rho})$ traceless	curvature (the stress tensor)
1^{+-}	$\text{Tr}(F[D,F])$	derivative structure

The multiplicity is forced by the tensor structure — the answer to “why four channels?” Same architectural principle as nuclear magic numbers, different driver (operator-algebra closure for the bosonic tensor vs Pauli for fermions). The odd-balls (0^{+-} , 1^{-+} , 2^{+-}) are forbidden at two gluons — clean unmixable channels.

7.2 The masses: one verified number sets the ladder

The glueball mass is the eigenvalue of the linear conformal Hamiltonian (the $\text{SO}(2)$ dilatation) on the holomorphic discrete series on $H^2(D_{IV}^5)$, diagonal in the K-type basis by Schur:

$m \propto E = \lambda_0 + (\text{energy step})$, $\lambda_0 = \text{genus}(D_{IV}^5) = n_C = 5$ (verified from the multiplicity formula $p = (r-1)a + b + 2 = 5$).

Energy step = $\text{SO}(5)$ harmonic degree = spin J ; the parity-odd (Hodge-dual) sector carries the half-canonical twist $n_C/2$. With one dimensionful anchor (seat = $\pi^5 \cdot m_e = 156.4 \text{ MeV}$; $m(0^{++}) = c_2 \cdot \text{seat} = 1720 \text{ MeV}$):

channel	step	ratio	substrate form	BST (MeV)	lattice (MeV)	dev	tier
0^{++}	0	1	—	1720	1730	−0.6%	anchor
2^{++}	$J = 2$	$7/5$	g/n_C	2408	2400	+0.3%	BLIND (genus + spin)
0^{-+}	$n_C/2$	$3/2$	N_c/rank	2580	2590	−0.4%	consistent (twist value-checked)
1^{+-}	$1 + n_C/2$	$17/10$	—	2924	2940	−0.5%	consistent (twist value-checked)

Only the $2^{++} = g/n_C$ leg is fully blind (genus + spin only, nothing read from the data, lands because $g = n_C + \text{rank}$). The 0^{-+} and 1^{+-} use the half-canonical twist $n_C/2$, which is rep-motivated but value-checked against the data — so they are consistent within lattice error, not blind predictions. (Quantization note: the *ratios*

live on the rung ladder $\lambda_0 = n_C$; the absolute scale comes only from anchoring 0^{++} at seat 11.)

7.3 The radial direction is linear, by Schur

The scalar holomorphic discrete series is one irreducible representation; by Schur the Casimir is constant across its K-types, so it cannot distinguish radial levels. The first radial scalar excitation $0^{++*} = (1,1)$ sits at the linear conformal energy $E = \lambda_0 + 2 = 7$, degenerate with 2^{++} (2408 MeV; lattice ~ 2670 , within quenched excited-gluon error). Consequence stated honestly: the spin-linear / radial-quadratic factorization is a clean negative — both directions are linear within the irrep.

7.4 Experimental and decay-side notes

The experimental scalar-gluon candidate $f_0(1710) \approx 1704$ MeV is consistent with BST's 0^{++} at seat 11 (1720 MeV) — the expected lightest-gluon match (not a unique identification; the scalar sector mixes with $q\bar{q}$). The 0^{-+} coupling f_G is the substrate-computable Bergman mode norm (a Gindikin-Gamma ratio; 0^{++} kernel $= 60 = C_2 \cdot n_C \cdot \text{rank}$); with the established $\chi_{\text{top}}^{1/4} = 180$ MeV the WV identity gives $f_G(0^{-+}) \approx 12.6$ MeV. Decay-side: mixing (gluon $\leftrightarrow q\bar{q}$ overlap on the shared Hardy space), not a “dump.”

7.5 Honest disposition of Section 7

The cross-channel spectrum is computed (four channels; one blind leg $2^{++} = g/n_C$; the others consistent within lattice error; multiplicity explained; radial direction settled by Schur; lightest state consistent with $f_0(1710)$). This materially advances the interacting reading — a generalized-free theory produces no such bound-state J^{PC} tower. It does not close the interacting-vs-free residual (Section 6). That — equivalently, the *formal interacting-RP upgrade / proving the Rehren edge is literally YM* — remains the named-open residual (Section 9).

8. The $so(7)$ unification (LEAD-STRENGTHENED)

Color and the Yang–Mills spectrum live in one algebra: $su(3) \subset g_2 \subset so(7)$, and $so(7)$ is the compact dual isometry of Q^5 — the same $so(7)$ whose Casimir spectrum is the gluon tower of Section 2; $g = 7$ is the $so(7)$ vector $= g_2$ fundamental $= 3 \oplus \bar{3} \oplus 1$. Color is not a geometric isometry of the noncompact domain ($su(3)$ not-subset-of $so(5,2)$) but is a geometric subalgebra of the compact-dual $so(7)$; on the domain side it is the operator algebra on H^2 . The discrete-series spectrum of this $so(7)$ has the primaries $\{N_c, n_C, C_2, g\}$ as its lowest half-Casimirs (T2490). Tier: LEAD-STRENGTHENED — the algebraic picture is whole; the bilinear-Schwinger realization on H^2 is in progress (#418).

9. Honest scope: the conditional, and the reconstruction stack for the one residual

We do not claim to have proved the Yang–Mills mass gap. The honest statement:

Given the reconstruction stack executes to rigor — including the interacting upgrade that identifies the edge theory as literally $SU(3)$ YM — the physical Yang–Mills theory exists as a rigorous 4D QFT with a mass gap whose derived compact-boundary value is $\text{mass}^2 = 2N_c = 6$ (color-rooted, R-independent, K1711), up to the $\Delta \rightarrow \text{mass}^2$ shift.

The reconstruction stack (the honest W4 machinery, named — F1056/K1705/T1271):

Hua-1963 boundary values $\rightarrow$ Bisognano–Wichmann-1975 / Borchers-2000 modular geometry $\rightarrow$ Rehren-2000 holographic net $\rightarrow$ Osterwalder–Schrader reconstruction $\rightarrow$ cluster decomposition.

Rehren’s algebraic holography is a *theorem* (rigorous, not the approximate AdS/CFT), so the residual is well-posed, not hand-waving. But it has two named, unfinished parts: (1) run the correspondence on D_{IV}^5 (build the bulk net and obtain the 4D boundary net); (2) identify the reconstructed boundary theory with the physical $SU(3)$ YM — the known subtlety that a Rehren/OS dual is not *automatically* the physical CFT. Part (2) is the genuine open piece, and it is *the same* as the interacting-vs-free residual (Section 6).

What is SOLID independent of the conditions: the gap exists and is nonzero (bounded $\Rightarrow$ gap) with no free dimensionless parameter beyond the ruler R , and its compact-boundary value is DERIVED: $\text{mass}^2 = 2N_c = 6$, color-rooted, R-independent (Section 2, K1711 — modulo residual (a) the $\Delta \rightarrow \text{mass}^2$ shift); the substrate-derivation of $SU(3)$ (Paper B); the OS *data* (Section 4); the so(7) unification (Section 8). The glueball 1720 MeV at 0.6% is a real, separate forward prediction (built on C_2^{int} , not the gap $2N_c$) but out of the room for the value-pin. What is materially advanced but still named-open: the cross-channel spectrum (Section 7). The flat-space residual (b) is the interacting-upgrade / literally-YM identification (Sections 6/9). This is a sharpening of the Clay problem to a derived compact-boundary value plus two named residuals — *not* a claim of its solution, never “YM proved.”

Explanatory summary (Guard 1 + Guard 2 held). BST *reproduces the scale and channel-structure of the Yang–Mills mass gap on a bounded domain*: the gap $\Delta = C_2 = 6$ is forced by representation theory (Guard 1: a domain operator, not the $\mathbb{R}^4$ Hamiltonian), its pure-glue reading matches the glueball at 0.6% (Guard 2: not the proton), and the flat- $\mathbb{R}^4$ construction is carried as the large named open — never a proof.

10. Open items and falsifiers

- Residual (a) — the $\Delta \rightarrow \text{mass}^2$ shift (Elie, K1711): the gap value is closed on the compact boundary (form (1,0), $\text{mass}^2 = 2N_c = 6$, R-independent). The

one remaining boundary-side question is whether the conformal weight Δ shifts the Laplacian-eigenvalue $\rightarrow$ physical-mass² identification, or mass² = 6 directly (which would harden the close). (*The Round-15/16 “three opens” are resolved: reading the field as a form fixes the operator, keeps the pure-spatial (1,0) mode, and removes the R-dependence — so (ii) and (iii) are settled; (i) is this Δ -shift.*)

- The one flat-space residual (Section 6/9): the interacting-upgrade / proving the Rehren-OS edge theory is literally SU(3) YM.
- #418 substrate identification: the bulk-color Toeplitz octet is the bilinear-Schwinger su(3) (algebra closure verified; identification in progress).
- Polyakov/Wilson-line mode at 10 vs deconfinement (Elie, stretch): the holonomy mode sits at 10, above the gap $2N_c = 6$ — a candidate tie to the confirmed deconfinement T_c . Check the ordering and the T_c relation.
- D_A dynamical-gap check (Elie): does the fluctuated Dirac D_A^2 carry the same gauge-form gap dynamically (mass² = $2N_c$ on the real boundary; state the space first)? Report “the D_{IV}^5 gap”; do not relabel $2N_c$ as “ C_2 .”
- Falsifier: the cross-channel glueball spectrum is a forward prediction — 0^{++} degenerate with 2^{++} at 2408 MeV is testable; if the cross-channel ratios fail, the interacting reading fails (the derived gap value $2N_c = 6$ and Paper B survive).

Count HOLDS 4 of 26. SU(3) scope; glueball masses are predictions, not SM parameter reductions. Paper A v0.9 (K1711 — the gap value CLOSED): reading the field as a form (the gauge field is the adjoint 1-form) fixes the operator as the form (Hodge) Laplacian, whose lowest S^4 eigenvalue is $2(n_C - 2) = 2N_c = 6$ (color-rooted, dimension-specific); the form parity keeps the pure-spatial (1,0) mode and gauge invariance eats A_θ , so mass² = $2N_c = 6$, R-independent — a derived value on the compact boundary. Label discipline (Guard 3): it is $2N_c$ (color), never “= C_2 ” (the scalar Casimir; the morning’s identity, K1707). The scalar readings (4, the (1,1) §570 candidate) were the wrong sector, superseded. Two named residuals: (a) the $\Delta \rightarrow$ mass² shift (Elie); (b) the flat- $\mathbb{R}^4$ construction (Clay). The Polyakov mode at 10 sits above the gap (deconfinement tie). The glueball 1720 stays separate (on C_2^{int}) — out of the room. The package intro can now move — the value is closed. NOT “YM proved” (residual (b) is the large $\mathbb{R}^4$ open), NOT “= C_2 .” INTERNAL. Nothing pushed; CP existence-only.

Draft v0.9 (K1711 — value closed). Spine: the field-is-a-form reading (gauge = adjoint 1-form $\rightarrow$ form Hodge Laplacian $\rightarrow 2N_c = 6$, color-rooted, dimension-generic, R-independent via A_θ gauge-eaten); Grace’s form- $\mathbb{Z}_2$ / dimension-genericity pin; Elie’s transverse-mode close + the $\Delta \rightarrow$ mass² residual + the Polyakov-10 deconfinement tie; Lyra’s write-up + label discipline ($2N_c \neq C_2$); the so(7) unification; Paper B (gauge-group derivation); Cal K1711 ($2N_c$ label, two residuals) + Keeper K1711 (gap close ruled, mis-assignment corrected). The scalar readings (Rounds 15/16) were the wrong sector — superseded. Companion to Paper B v0.6.

— Lyra, Wed 2026-08-19 (date-verified). Paper A v0.9, K1711: gap value CLOSED —

$\text{mass}^2 = 2N_c = 6$, color-rooted, R-independent, derived on the compact boundary
 (gauge field = form $\rightarrow$ Hodge Laplacian $\rightarrow 2N_c$; A_θ gauge-eaten $\rightarrow$ pure-spatial
 (1,0)). Labeled $2N_c$, never “ $= C_2$ ”. Two residuals: $\Delta \rightarrow \text{mass}^2$ shift, flat- $\mathbb{R}^4$ construc-
 tion. Glueball separate-valid, out of the room. Package intro can move.